



Lab 05 - Object deserialization from text

A. Exercises

Create a solution named **Week05**

1. IntegersText

- Read all integers from `input.txt` file.

```
4 17 18 19 20
```

- Find all odd numbers, all even numbers, all palindromic numbers, all prime numbers, all square numbers, all perfect numbers
- Find the minimum, maximum, average value (decimal number), sum of all numbers

Sample output

```
Working with integer arrays from text files Reading data from input.txt.  
Expected to find 4 numbers. Found 4 numbers: 17 18 19 20 Found 2 odd numbers:  
17, 19 Found 2 even numbers: 18, 20 Found 0 palindromic number Found 2 prime  
numbers: 17, 19 Found 0 square number Found 0 perfect number Minimum value:  
17 Maximum value: 20 Average value: 18.5 Sum of all numbers: 74 Program is  
exiting. Press enter to quit...
```

Hints

```
class IntegersTextProvider { public: static vector<int> read(std::string filename); };
```

To convert a string into an integer

```
class Integer { public: static int parse(std::string info); };
```

2. FractionsText

- Read all fractions from an `input.txt` file

```
4 8/3 6/4 2/1 15/17
```

- Print out all the fractions that you get from the files
- Print out all the fractions in their lowest term
- Print out all the fractions in their decimal form
- Print out all the fractions in their percentage form

Sample output

```
Working with fraction arrays from text files Reading data from input.txt.  
Expected to find 4 fractions. Found 4 fractions: 8/3, 6/4, 2/1, 15/17 Lowest  
term: 2 2/3, 3/2, 2, 15/17 Decimal: 2.66, 1.50, 2.00, 0.88 Percentage: 266%,  
150%, 200%, 88% Program is exiting. Press enter to quit...
```

B. Submission instructions

1. Delete the hidden folder **.vs**
2. Compress into the format {StudentID}.zip/.rar and submit back this file.
 - For example if your student id is 1812445, your filename should be 1812445.zip or 1812445.rar

C. Frequently asked questions

No question yet.